Program : **Diploma in Renewable Energy**	
Course Code : **6299A**	Course Title: **Power Quality Lab**
Semester : **6**	Credits: **1.5**
Course Category: **Program Elective**	
Periods per week: **3 (L: 0 T: 0 P: 3)**	Periods per semester: **45**

Course Objectives:

- To understand the voltage sag characteristics and it's severity
- To know the basic concept on power compensation.
- To introduce the concept of harmonic analysis.
- To impart a basic knowledge on power quality.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Concept of current flow through various loads in converters and inverters.		**Fundamentals of Power Electronics**	**5**

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Identify the characteristics of voltage sag and to estimate it's severity.	12	Applying
CO2	Perform simulations for power compensation and power angle curve.	9	Applying
CO3	Make use of simulations to understand the concepts of harmonic analysis.	12	Applying
CO4	Develop the power quality monitoring model.	6	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	**Identify the characteristics of voltage sag and to estimate it's severity**.		
M1.01	Familiarize the basic tools in MATLAB or similar SOFTWARE.	3	Understanding
M1.02	Perform basic simulations in MATLAB/simulink/ or Similar Software	3	Applying
M1.03	Develop the voltage sag characteristics using simulation.	3	Applying
M1.04	Perform a test to estimate sag severity in overhead conductors.	3	Applying
CO2	**Perform simulations for power compensation and power angle curve.**		
M2.01	Perform a test to simulate power factor correction	3	Applying
M2.02	Plot a set of power angle curve for a salient pole synchronous generator using simulation.	3	Applying

M2.03	Construct a circuit and perform compensation of active and reactive power using simulation.	3	Applying
	Lab Test – I	3	
CO3	**Make use of simulations to understand the concepts of harmonic analysis.**		
M3.01	Model a circuit to determine Voltage distortion using MATLAB	3	Applying
M3.02	Model a circuit to know the impact of harmonic distortion.	3	Applying
M3.03	Perform a test on harmonic analysis of power system	3	Applying
M3.04	Perform a study on power system response characteristics	3	Applying
C04	**Develop the power quality monitoring model.**		
M4.02	Perform a test to simulate power quality analysis.	3	Applying
M4.04	Construct a model of DSTATCOM using MATLAB SIMULINK or similar softwares	3	Applying
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
T1	Power Quality in Electrical Systems/Alexander Kusko & Marc T Thompson
T2	Electrical Power System Quality/Roger C Dugan/Mc Graw Publications
R1	Power Quality in modern power systems/Sanjeev Kumar Padmanabhan
R2	Power Quality/Andreas Eberhand

Online Resources

Sl.No	Website Link
1	https://www.mathworks.com/help/physmod/sps/examples/d-statcom-average-model.html
2	https://www.primo-engineering.com/2020/12/write-matlab-script-to-plot-set-of-per.html?m=1
3	https://in.mathworks.com/solutions/electrification/power-factor-correction.html

Open Ended Experiment

1. **Perform a test to model and simulate power quality disturbances.**